EXP 7. Breadth-First Search (BFS)

Aim

To implement Breadth-First Search to traverse a graph level by level.

Algorithm

1. Create the graph using an adjacency list.
2. Add the starting node to a queue and mark it visited.
3. Remove a node from the front of the queue and print it.
4. Add each unvisited neighbour to the queue and mark it visited.
5. Repeat until the queue is empty.

Program

*AI.py - C:/Users/Barath/AppData/Local/Programs/Python/Pythc

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```
from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": [],
    "F": []
}

def bfs(start):
    visited = {start}
    q = deque([start])

    while q:
        node = q.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                q.append(neighbour)

bfs("A")
```

Output

```
>>
    ====================================================
    A B C D E F
>>
```

Result

Thus, BFS traversed the graph level by level from node A.